

CS57300
PURDUE UNIVERSITY

DATA MINING

LOGISTICS

- ▶ Assignment 1 is on Gradescope
 - ▶ Due on September 3 (Sunday), 11:59pm
 - ▶ Let TAs know immediately if you are not on Gradescope
- ▶ TA office hours:
 - ▶ Swagarika Giri (giri2@purdue.edu): Wednesdays 4-5pm (HAAS 143)
 - ▶ Weixian Zhang (zhan2156@purdue.edu): Thursdays 3-4pm (HAAS 143)
 - ▶ Hasan Amin Mahmood (mahmood6@purdue.edu): Friday 11am-noon (HAAS 143)

POPULATIONS AND SAMPLES

ELEMENTARY UNITS, POPULATIONS, AND SAMPLES

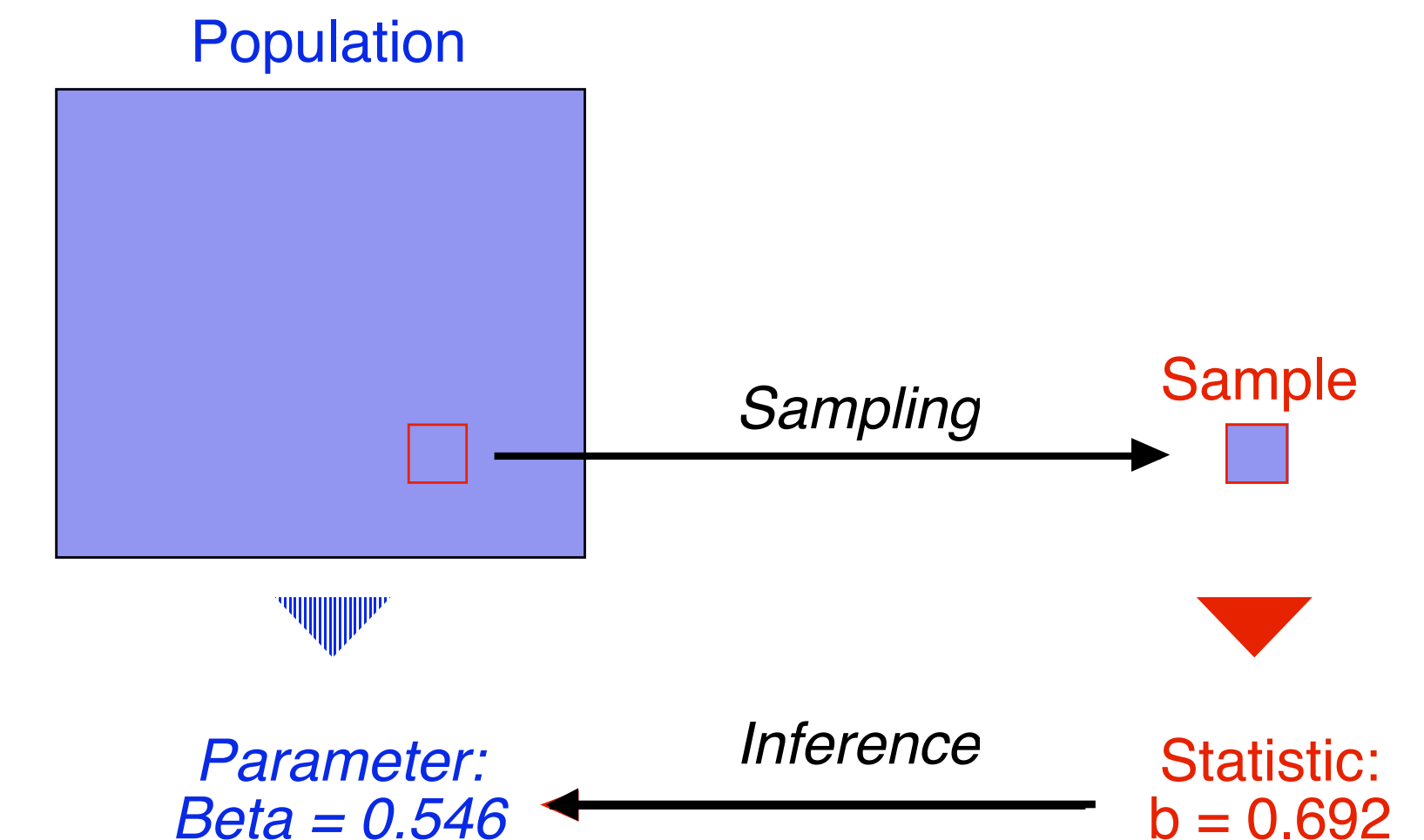
- ▶ Elementary units:
 - ▶ Entities (e.g., persons, objects, events) that meet a set of specified criteria
 - ▶ Example: A person who has purchased something from Walmart in the past month
- ▶ Population:
 - ▶ Aggregate of elementary units (i.e, all entities of interest)
- ▶ Sample:
 - ▶ Sub-group of the population

SAMPLING

- ▶ Reasons to sample
 - ▶ Obtaining the entire set of data of interest is too expensive or time consuming
 - ▶ Processing the entire set of data of interest is too expensive or time consuming
- ▶ Sampling is the main technique employed for data selection

USE SAMPLES FOR ESTIMATION

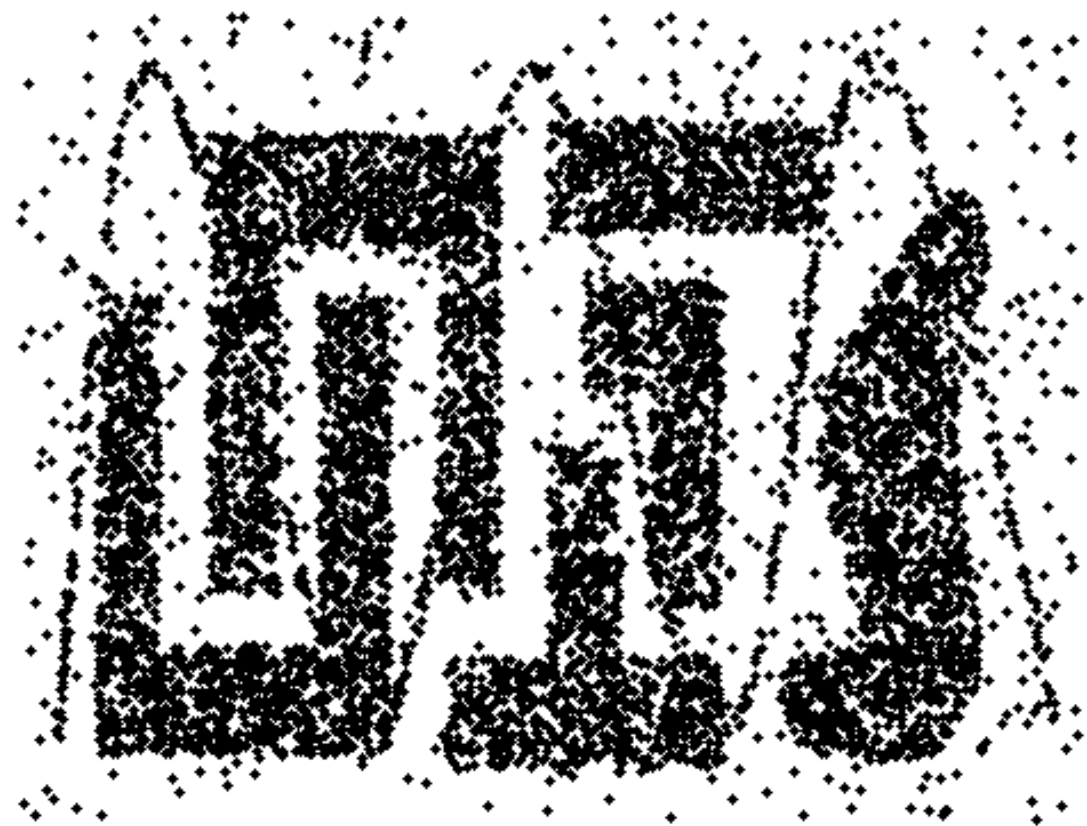
- ▶ In data mining we often work with a sample of data from the population of interest
- ▶ If we had the population we could calculate the properties of interest
- ▶ Sample serves as a reference group for **estimating** characteristics about the population and drawing conclusions



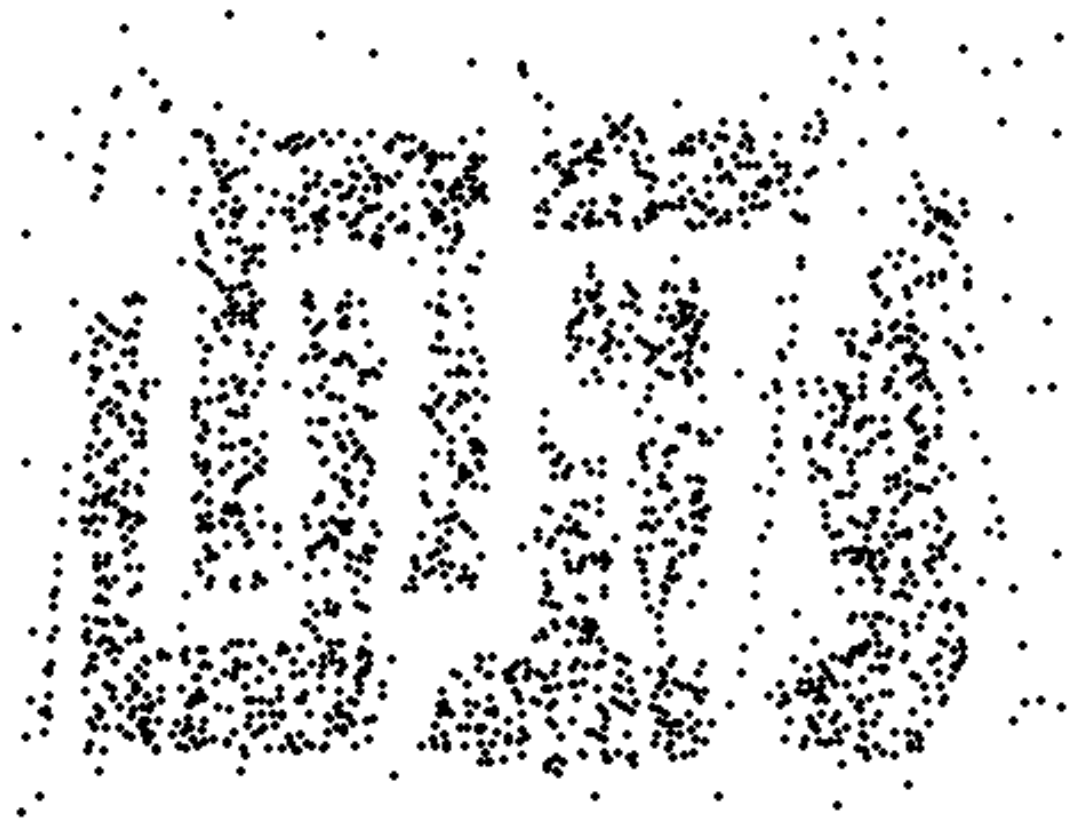
PRINCIPLE FOR EFFECTIVE SAMPLING

- ▶ The key principle for effective sampling is the following:
 - ▶ Using a sample will work almost as well as using the entire data set, if the sample is **representative**
 - ▶ A sample is representative if it has approximately the same property (of interest) as the original set of data

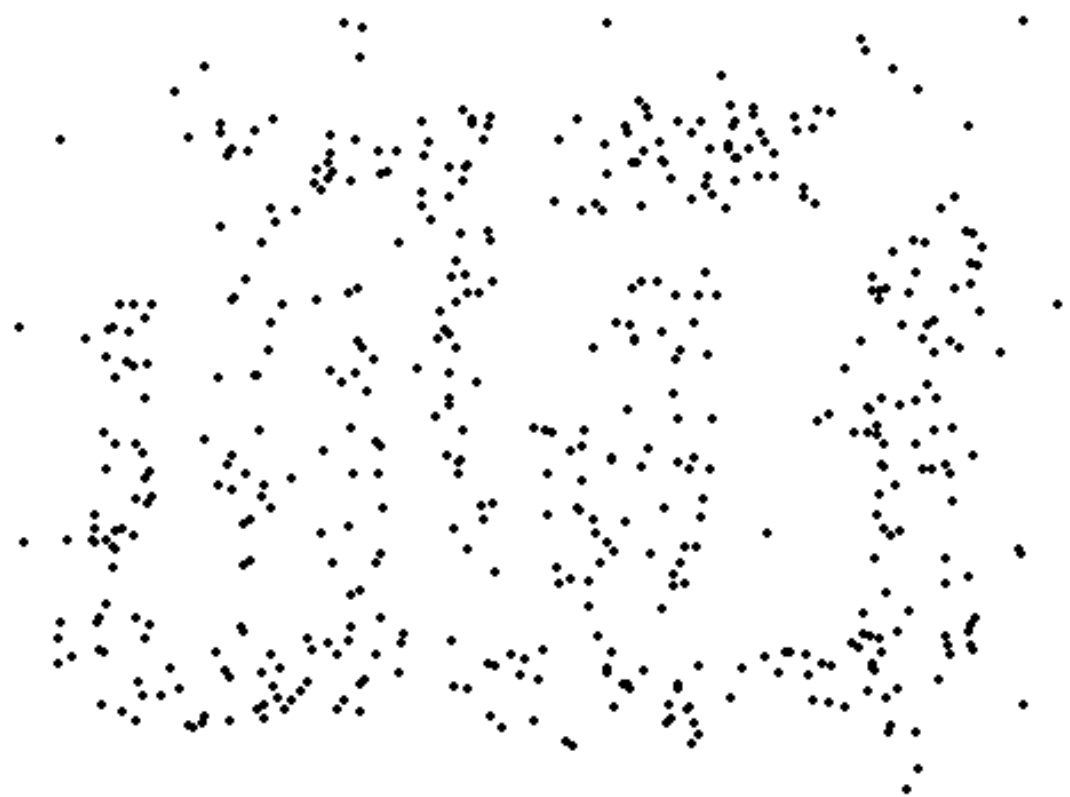
SAMPLE SIZE



8000 Points



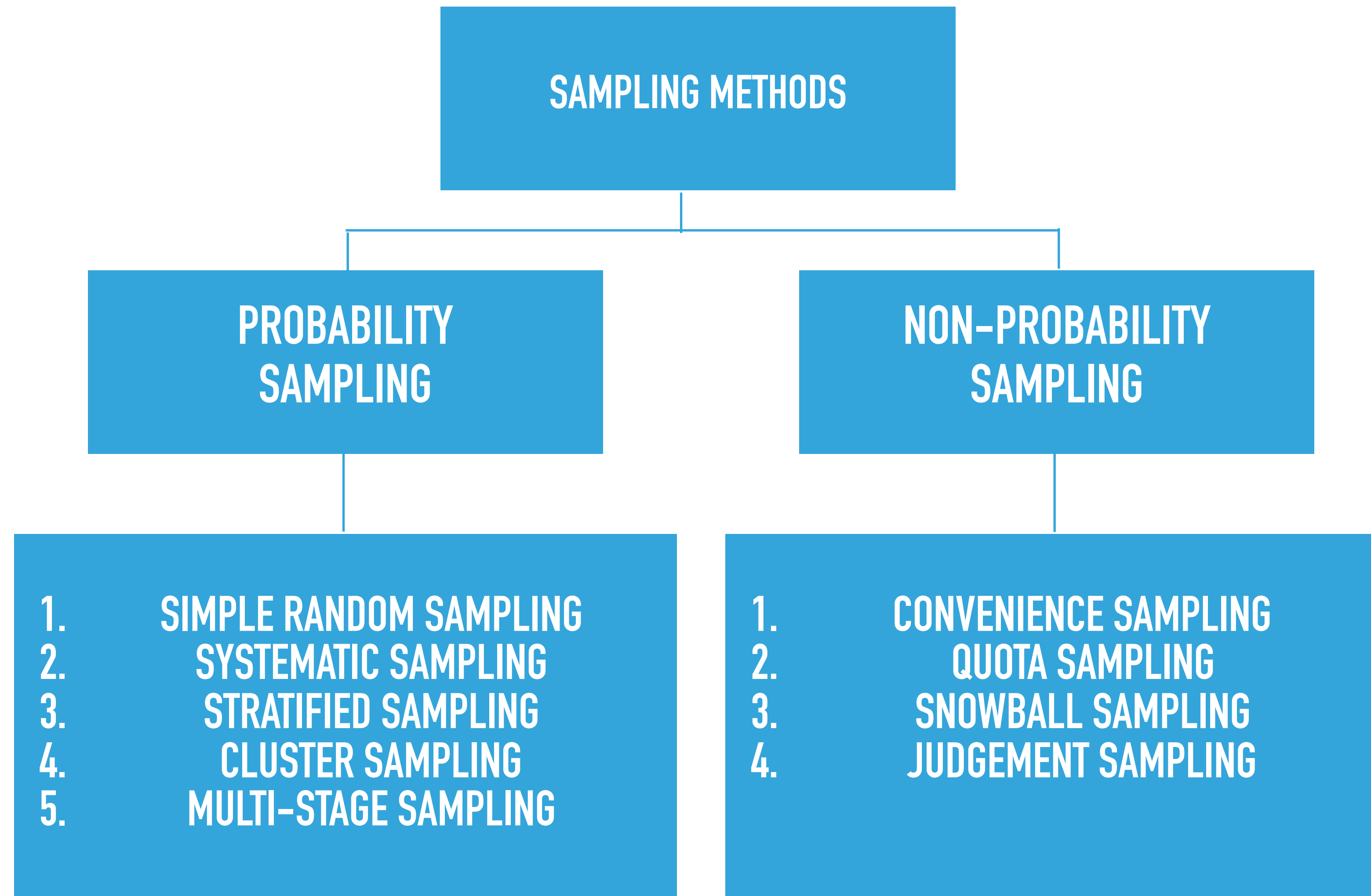
2000 Points



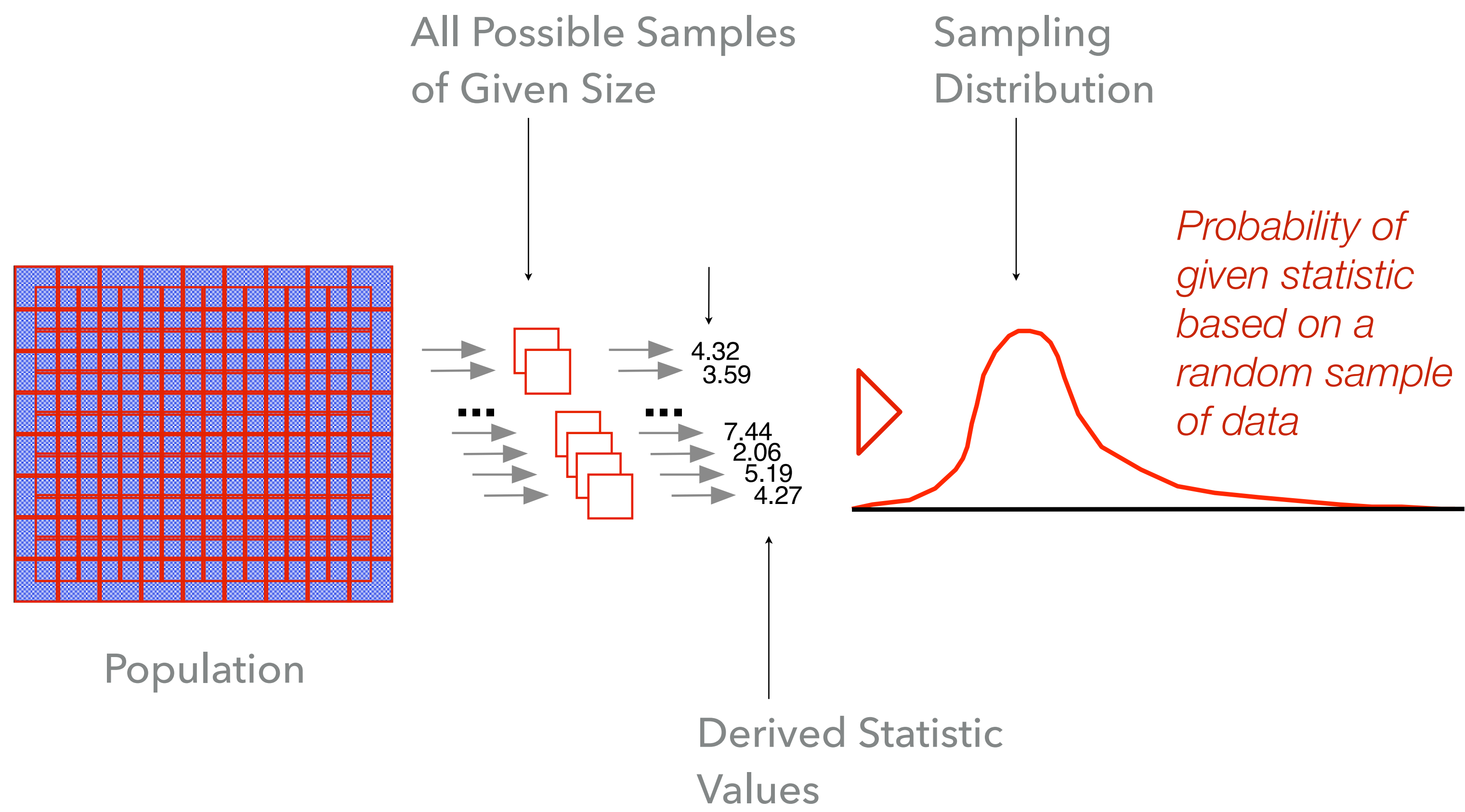
500 Points

TYPES OF PROBABILITY SAMPLING

- ▶ Simple random sampling
 - ▶ There is an equal probability of selecting any particular item
 - ▶ Sampling without replacement
 - ▶ As each item is selected, it is removed from the population
 - ▶ Sampling with replacement
 - ▶ Items are not removed from the population as they are selected for the sample; the same item can be picked up more than once
- ▶ Stratified sampling
 - ▶ Split the data into several partitions; then draw random samples from each partition



SAMPLING DISTRIBUTIONS



STATISTICAL INFERENCE

STATISTICAL INFERENCE

- ▶ Infer properties of an unknown distribution with sample data generated from that distribution
- ▶ Parameter estimation
 - ▶ Infer the value of a population parameter based on a sample statistic (e.g., estimate the mean)
- ▶ Hypothesis testing
 - ▶ Infer the answer to a question about a population parameter based on a sample statistic (e.g., is the mean non-zero?)

PARAMETER ESTIMATION

- ▶ Infer the value of population parameters (θ) from data
- ▶ θ can take values in the parameter space Θ
- ▶ Frequentist approach
 - ▶ Population parameters are fixed but unknown
 - ▶ Data is a random sample drawn from population
 - ▶ Use maximum likelihood estimation (MLE)
- ▶ Bayesian approach
 - ▶ Parameters are random variables with a distribution of possible values
 - ▶ Data is fixed and known, provides evidence for different parameter values
 - ▶ Use maximum a posteriori estimation (MAP)

MAXIMUM LIKELIHOOD ESTIMATION (MLE)

- ▶ Suppose we have a set of data $X = \{x_i\}_{i=1}^N$ independently drawn from the population
- ▶ The maximum likelihood estimation finds the parameter values that maximize the likelihood of observing the data

$$\theta_{MLE} = \arg \max_{\theta} P(X|\theta)$$

$$= \arg \max_{\theta} \prod_i P(x_i|\theta)$$

$$= \arg \max_{\theta} \log \prod_i P(x_i|\theta)$$

$$= \arg \max_{\theta} \sum_i \log P(x_i|\theta)$$

MAXIMUM A-POSTERIORI ESTIMATION (MAP)

- ▶ Suppose we have a set of data $X = \{x_i\}_{i=1}^N$ independently drawn from the population, and the prior distribution for the parameter is $P(\theta)$
- ▶ The maximum a-posteriori estimation finds the mode of the posterior distribution of the parameters

$$\theta_{MAP} = \arg \max_{\theta} P(X|\theta)P(\theta)$$

$$= \arg \max_{\theta} \log P(X|\theta)P(\theta)$$

$$= \arg \max_{\theta} \log \prod_i P(x_i|\theta)P(\theta)$$

$$= \arg \max_{\theta} \sum_i \log P(x_i|\theta)P(\theta)$$

MLE VS. MAP EXAMPLE

- ▶ Flip a coin for N times and observe n heads; what's the probability of seeing the head if tossing the coin once?
- ▶ Likelihood of observing the data: $P(D | \theta) = \binom{N}{n} \theta^n (1 - \theta)^{N-n}$
 - ▶ The number of heads observed follows a binomial distribution
- ▶ Maximum likelihood estimation:

$$\theta_{MLE} = \operatorname{argmax}_{\theta} P(D | \theta) = \frac{n}{N}$$

MLE VS. MAP EXAMPLE

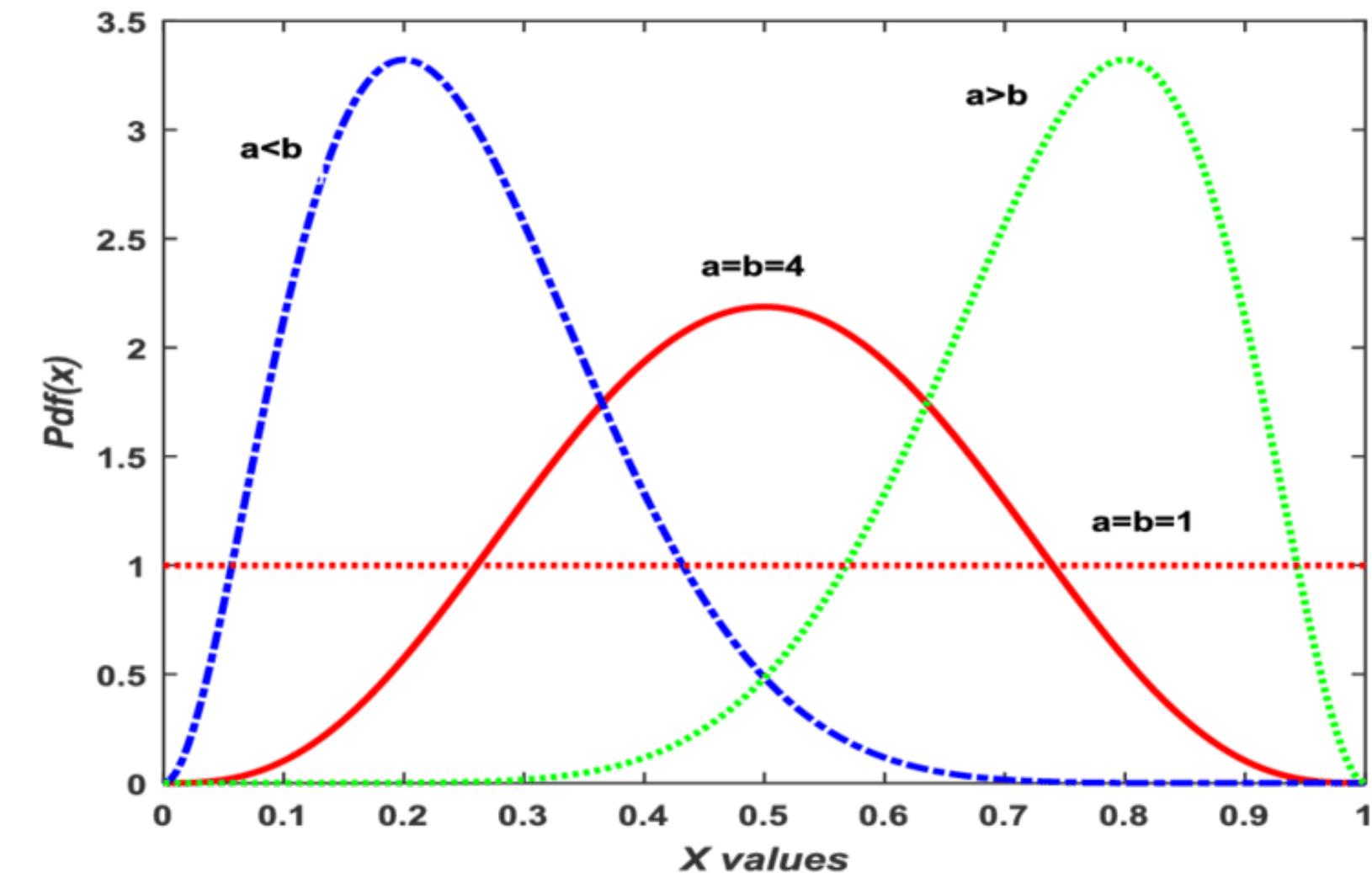
- ▶ Maximum a-posteriori estimation:
- ▶ Suppose the prior is a Beta distribution

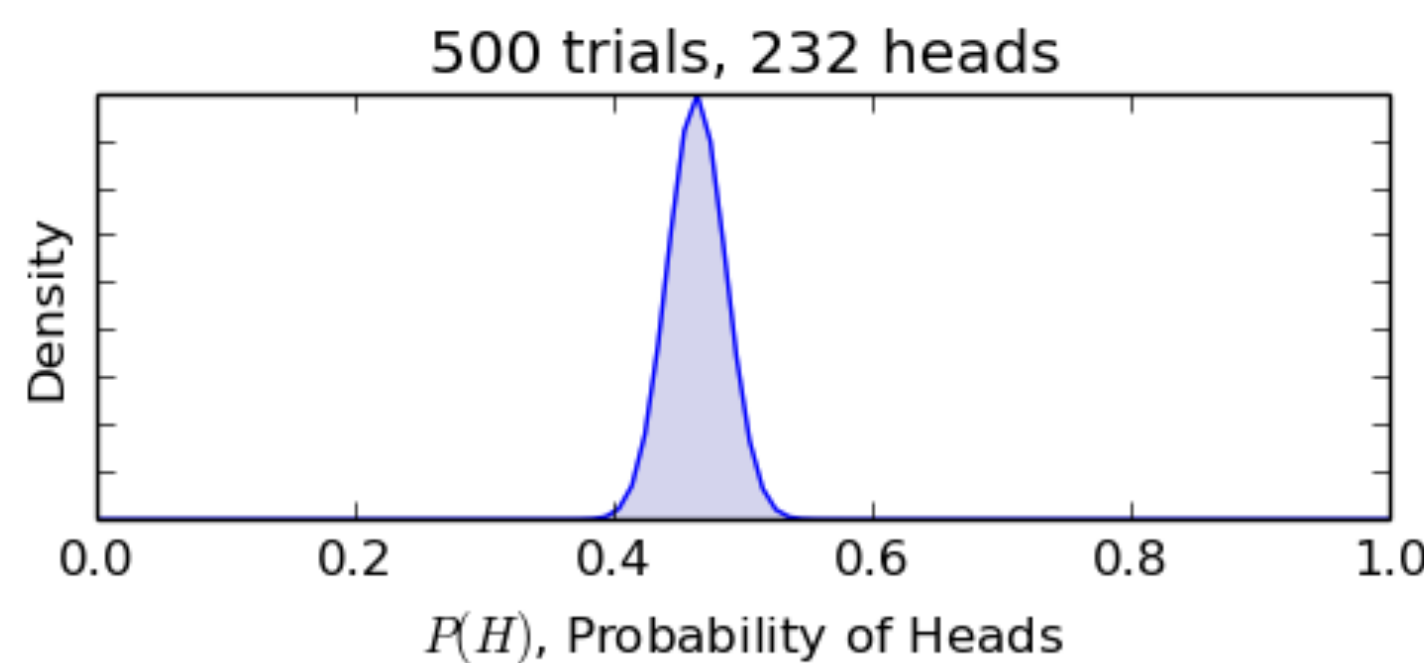
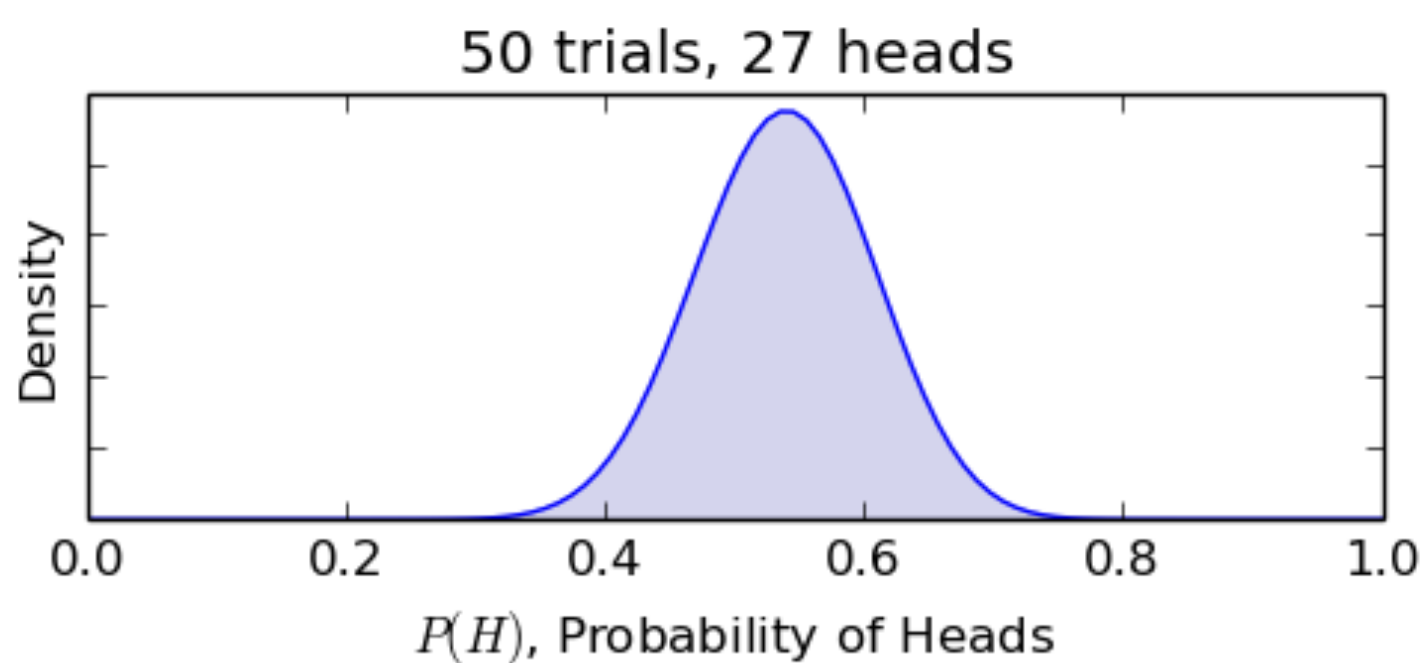
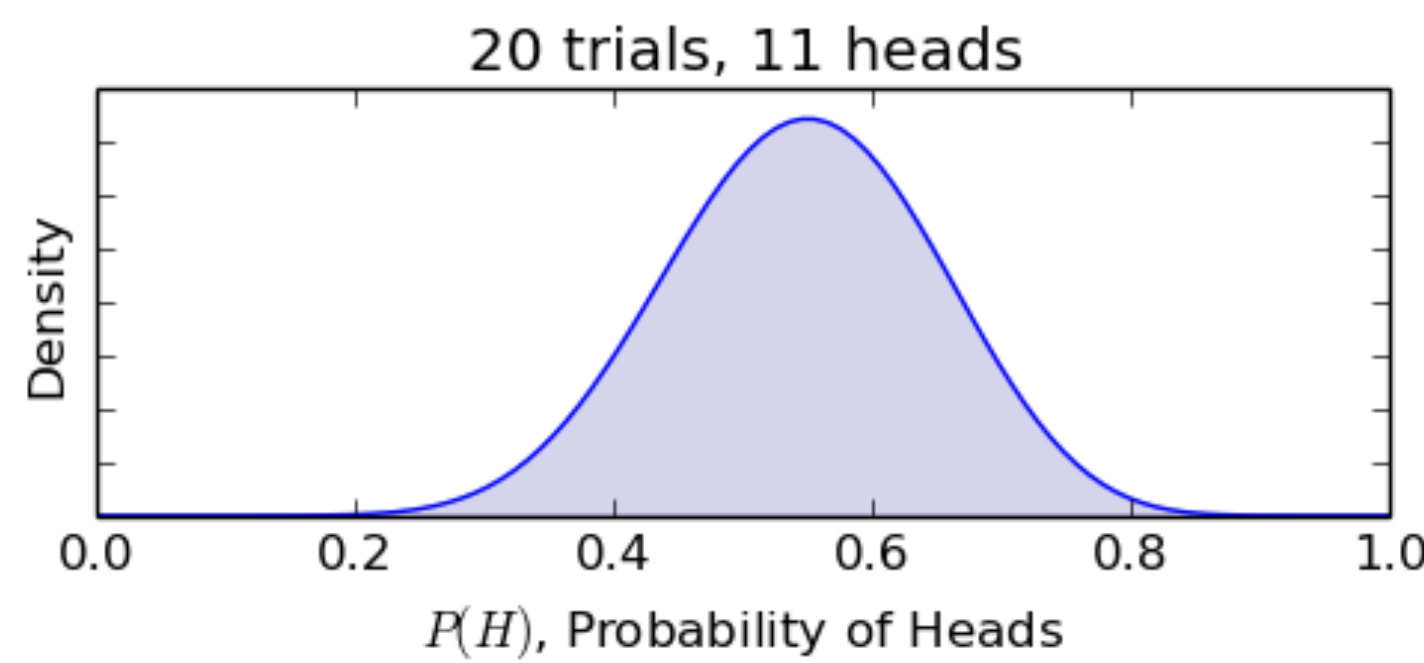
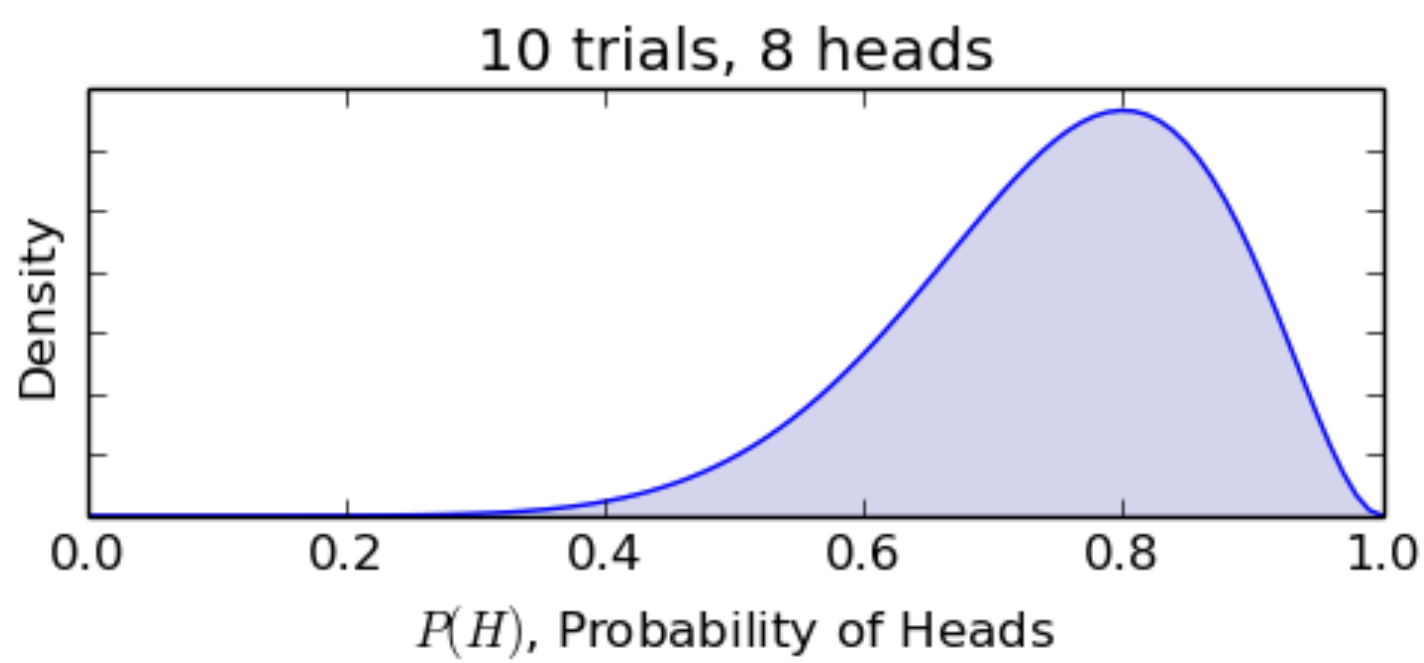
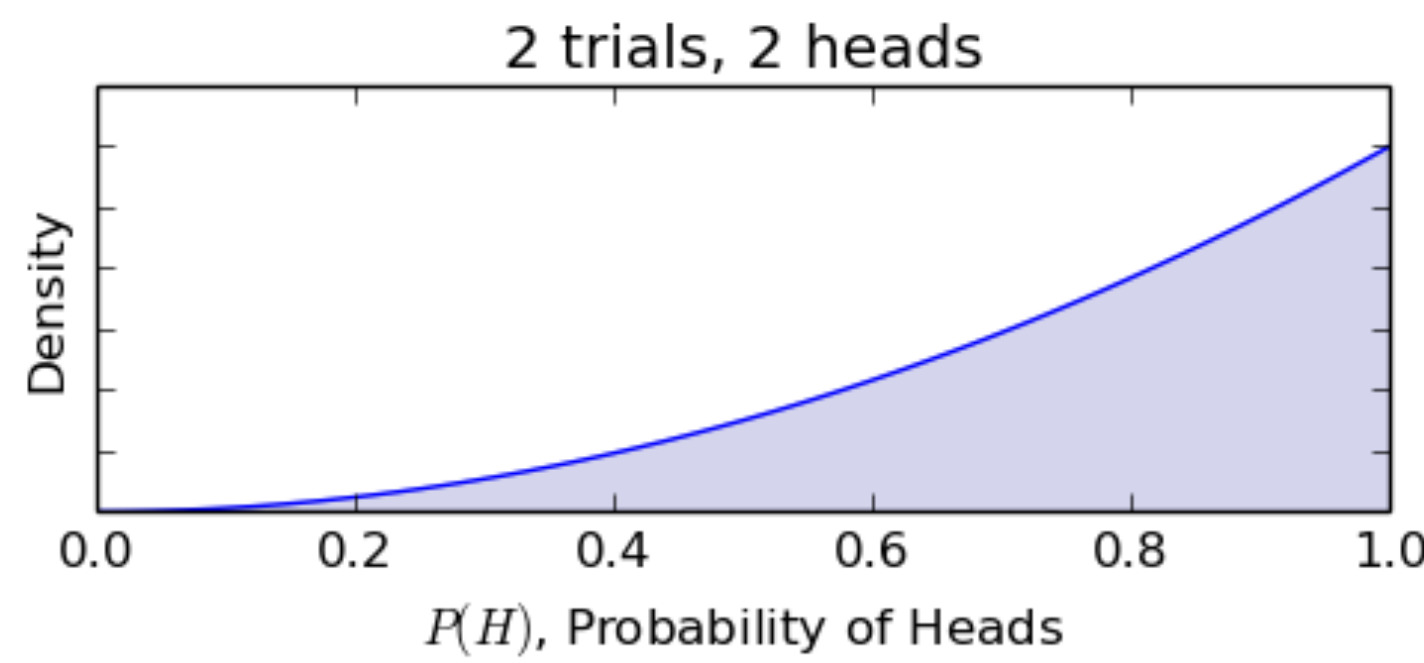
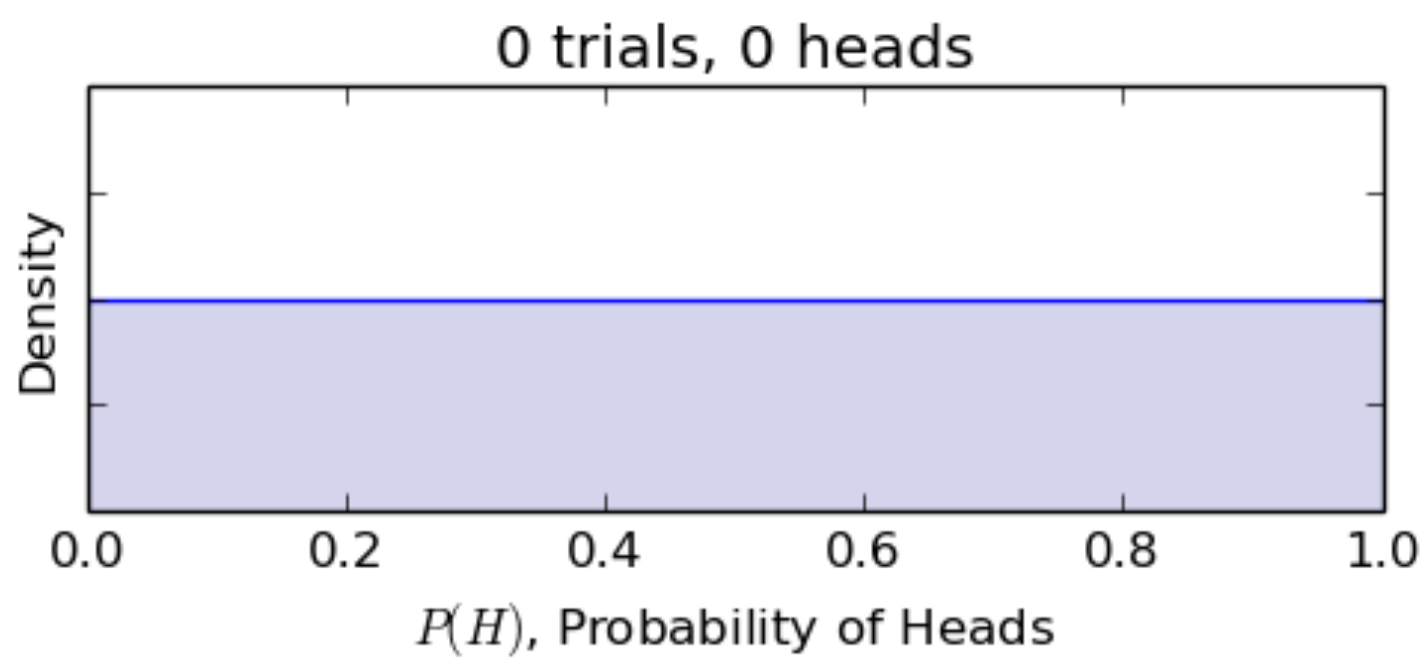
$$P(\theta) = \frac{\theta^{a-1}(1-\theta)^{b-1}}{B(a,b)} \sim \text{Beta}(a,b), \text{ where } B(a,b) = \int_0^1 \theta^{a-1}(1-\theta)^{b-1} d\theta = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}, \Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$$

- ▶ Then, the posterior is:

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} = \frac{\binom{N}{n}(\theta^{a+n-1}(1-\theta)^{b+N-n-1}/B(a,b))}{\int_0^1 \binom{N}{n}(\theta^{a+n-1}(1-\theta)^{b+N-n-1}/B(a,b))d\theta}$$

$$\sim \text{Beta}(a+n, b+N-n)$$





MLE VS. MAP EXAMPLE

- ▶ Maximum a-posteriori estimation:

$$\begin{aligned}\theta_{MAP} &= \operatorname{argmax}_{\theta} P(\theta | D) \\ &= \operatorname{argmax}_{\theta} \operatorname{Beta}(a + n, b + N - n) \\ &= \frac{a + n - 1}{a + b + N - 2}\end{aligned}$$

- ▶ Notice that in this example, the posterior distribution is in the same probability distribution family as the prior distribution.
 - ▶ The prior and posterior are called **conjugate distributions**, and the prior is called a **conjugate prior** for the likelihood function.
 - ▶ The beta distribution is a conjugate prior to the binomial likelihood.